

CHE 201 Fall 2019 HW 7

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TOTAL POINTS

84 / 100

QUESTION 1

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- **0 pts** Correct
- **1 pts** a) Wrong schematic (A2 should be $< A1$)
- **1 pts** a) wrong or missing system boundary
- **1.5 pts** c) Missing mass balance
- **4 pts** d) Wrong value, wrong calculation or missing T2. Correct answer is $h_2 = 99.5 \text{ Btu/lb}$ and $T_2 = -2.4 \text{ }^\circ\text{F}$
- **2 pts** e) State 1 is superheated refrigerant
- ✓ - **2 pts** e) State 2 is saturated refrigerant (L+V mixture)
- **4 pts** e) Wrong or missing P-v diagram

QUESTION 2

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- **0 pts** Correct
- ✓ - **1 pts** a) Wrong or missing schematic
- **1 pts** a) wrong or missing system boundary
- **2 pts** b) Wrong or missing eng model
- **1.5 pts** c) Missing mass balance
- **2 pts** d) Missing or wrong exit area. $A_2 = 0.06 \text{ m}^2$
- **4 pts** d) mass flow rate = 4.5 kg and $A_2 = 0.06 \text{ m}^2$
- **2 pts** e) $V_2 > V_1$ (expansion)
- ✓ - **4 pts** a) Missing or wrong T-v diagram

QUESTION 3

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- **0 pts** Correct
- ✓ - **1 pts** a) Wrong schematic
- **1 pts** a) Missing or wrong system boundary
- **2 pts** b) State 1 is water vapor
- **4 pts** b) Wrong or missing P-v diagram
- ✓ - **2 pts** b) $V_2 > V_1$ (expansion)
- **2 pts** b) State 2 is saturated vapor

- **2 pts** c) Missing eng model
- **3 pts** d) Missing simplified mass rate and energy rate balance.
- **1.5 pts** d) Missing simplified mass rate
- **4 pts** e) $W = 3581 \text{ kW}$

QUESTION 4

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- **0 pts** Correct
- ✓ - **3 pts** T-V figure
- ✓ - **1 pts** T-V figure, $v_1 = ?$ $v_2 = ?$
- **5 pts** Q calculation incorrect
- **3 pts** scheme incorrect, you mention $\Delta PE = 0$ but your input and output is not on the same level
- **1 pts** T-V figure where is step 1 and 2
- **3 pts** scheme incorrect. input? output?
- **5 pts** m incorrect
- **5 pts** scheme

QUESTION 5

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- ✓ - **0 pts** Correct
- **3 pts** scheme, storage tank that is 15 m above reservoir
- **3 pts** storage tank that is 15 m above reservoir; the pump and the storage tank should be on the same level
- **5 pts** w calculation incorrect
- **3 pts** unit in KW
- **3 pts** $h_1 = ?$ $h_2 = ?$
- **3 pts** engineering model

QUESTION 6

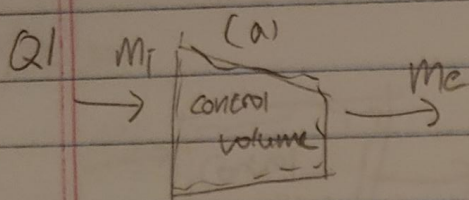
6 6 18 / 20

- **0 pts** Correct
- **1 pts** a) did not label the system boundary

- 1 pts a) wrong schematic
- 4 pts b) missing the engineering model
- ✓ - 2 pts c) missing the mass rate balance
- 2 pts d) missing or wrong energy rate balance
- 4 pts d) used wrong enthalpy values to solve for x
- 5 pts d) $x = 0.194$
- 2.5 pts e) State 1 is compressed liquid
- 5 pts e) wrong or missing sketch of the process on

a T-v diagram

- 2.5 pts e) $v_2 > v_1$
- 2.5 pts e) missing point 2 on the T-v diagram
- 2.5 pts e) $T_1 > T_2$



model (b)

1 inlet 1 outlet.

$$\Delta PE = 0, \dot{W} = 0, \frac{dE_{cv}}{dt} = 0 \text{ (Steady state)}, \dot{Q} = 0$$

control volume -

$$\dot{m}_i = \dot{m}_e = \dot{m}$$

$$\dot{Q} = \dot{m} [h_1 - h_2]$$

$$0 = \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} \right]$$

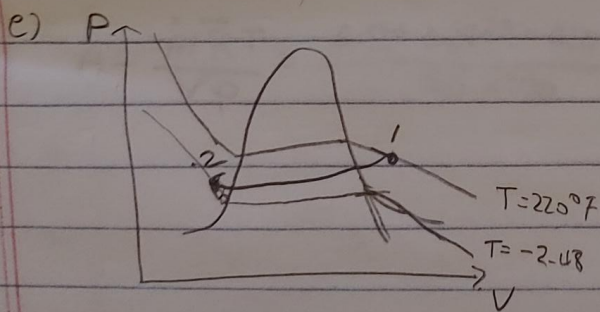
(d) $h_1 = 144.15 \text{ Btu/lb}$ $V_1 = 120 \text{ ft/s}$ $V_2 = 1500 \text{ ft/s}$

$$0 = \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} \right]$$

$$h_2 = h_1 + \frac{V_1^2 - V_2^2}{2}$$

$$= 144.15 \text{ Btu/lb} + 0.5 (120^2 - 1500^2) \frac{\text{ft}^2}{\text{s}^2} \times \frac{1 \text{ Btu/lb}}{25037 \text{ ft}^2/\text{s}^2} = 99.51 \text{ Btu/lb}$$

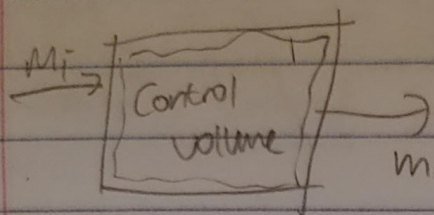
$$T_2 = -2.48^\circ \text{F}$$



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- 0 pts Correct
- 1 pts a) Wrong schematic (A2 should be < A1)
- 1 pts a) wrong or missing system boundary
- 1.5 pts c) Missing mass balance
- 4 pts d) Wrong value, wrong calculation or missing T2. Correct answer is $h_2 = 99.5 \text{ Btu/lb}$ and $T_2 = -2.4 \text{ oF}$
- 2 pts e) State 1 is superheated refrigerant
- ✓ - 2 pts e) State 2 is saturated refrigerant (L+V mixture)
- 4 pts e) Wrong or missing P-v diagram

Q2 (a)



(b) model

- steady state $m_i = m_e = \dot{m}$
- $2500 \text{ kW} = \dot{W}$ - $\dot{Q} = 0$, $\Delta PE = 0$, $\Delta KE = 0$
- 1 inlet & outlet
- ideal gas
- control volume

$$(c) 0 = -\dot{W}_{cv} + \dot{m}(h_1 - h_2)$$

$$\dot{W}_{cv} = \dot{m}$$

$$h_1 - h_2$$

$$h_1 = 1000.55 \text{ kJ/kg}$$

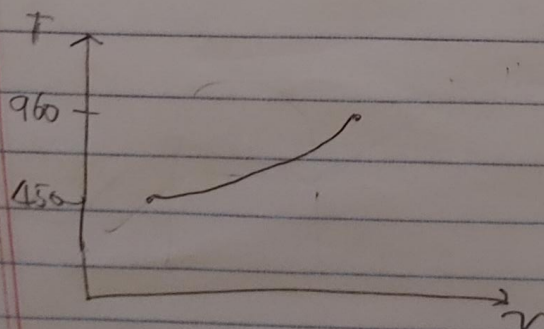
$$\dot{W}_{cv} = 2500 \text{ kJ/s}$$

$$h_2 = 451.80 \text{ kJ/kg}$$

$$\frac{2500}{1000.55 - 451.80} = \dot{m} = 4.55 \text{ kg/s}$$

$$\dot{m} = \frac{A \bar{v}}{v}, \quad \bar{v} = \frac{\bar{R} T}{P} \Rightarrow \dot{m} = \frac{A \bar{v} P}{\bar{R} T}$$

$$A = \frac{\dot{m} \bar{R} T}{\bar{v} P} = \frac{4.55 \times 0.2869 \times 450}{90 \times 100} = 0.0653 \text{ m}^2$$



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- 0 pts Correct

✓ - 1 pts a) Wrong or missing schematic

- 1 pts a) wrong or missing system boundary

- 2 pts b) Wrong or missing eng model

- 1.5 pts c) Missing mass balance

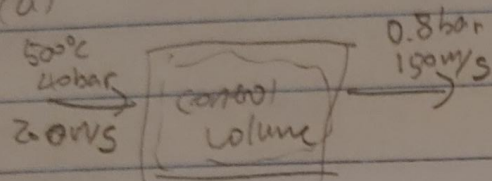
- 2 pts d) Missing or wrong exit area. $A_2 = 0.06 \text{ m}^2$

- 4 pts d) mass flow rate = 4.5 kg and $A_2 = 0.06 \text{ m}^2$

- 2 pts e) $V_2 > V_1$ (expansion)

✓ - 4 pts a) Missing or wrong T-v diagram

Q3 (a)

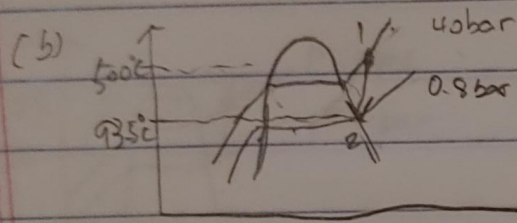


(b) model

- Control volume - adiabatic

- Steady state - water

- $\Delta PE = 0$ - $m_i = m_e$ - $Q = 0$



d) $0 = \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) \right] - \dot{W}$

e) $v_1 = 0.08643 \text{ m}^3/\text{kg}$ $h_1 = 3445.3 \text{ kJ/kg}$

$v_2 = 2.087 \text{ m}^3/\text{kg}$ $h_2 = 2665.8 \text{ kJ/kg}$

$\dot{m} = 9.48 \text{ m/s}$

$\frac{9.48 \text{ m/s}}{2.087 \text{ m}^3/\text{kg}} = 4.54 \text{ kg/s}$

$\dot{W} = \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) \right] = 4.54 \left[(3445.3 - 2665.8) + \left(\frac{200^2 - 150^2}{2} \right) \times \frac{1}{10^3} \right]$

$= 3578.65 \text{ kW}$

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- 0 pts Correct

✓ - 1 pts a) Wrong schematic

- 1 pts a) Missing or wrong system boundary

- 2 pts b) State 1 is water vapor

- 4 pts b) Wrong or missing P-v diagram

✓ - 2 pts b) $V_2 > V_1$ (expansion)

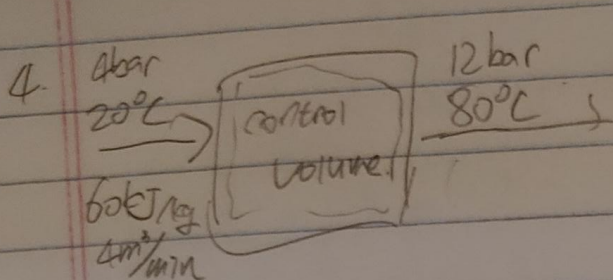
- 2 pts b) State 2 is saturated vapor

- 2 pts c) Missing eng model

- 3 pts d) Missing simplified mass rate and energy rate balance.

- 1.5 pts d) Missing simplified mass rate

- 4 pts e) $W = 3581 \text{ kW}$



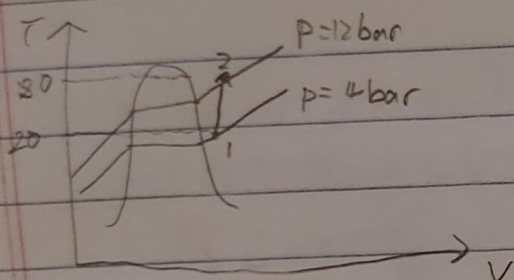
Model

- R-134A.

- Steady/State. - Control Volume

- inlet & outlet. - $m_i = m_e = m$

- $\Delta KE = 0, \Delta PE = 0$. - compressor.



(c) $0 = Q_{cv} - W_{cv} + \dot{m}[h_1 - h_2]$

(d) $Q_{cv} = W_{cv} - \dot{m}[h_1 - h_2]$ $h_1 = 265.86 \text{ kJ/kg}$ $h_2 = 311.39 \text{ kJ/kg}$

$\dot{m} = \frac{AV}{v} = \frac{4 \text{ m}^3}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ s}} \times \frac{1 \text{ kg}}{0.054213 \text{ m}^3} \quad v_1 = 0.054213 \text{ m}^3/\text{kg}$

$= 1.229 \frac{\text{kg}}{\text{s}}$

$Q_{cv} = \frac{-60 \text{ kJ}}{\text{kg}} - 1.229 \frac{\text{kg}}{\text{s}} \left[(265.86 - 311.39) \frac{\text{kJ}}{\text{kg}} \right]$

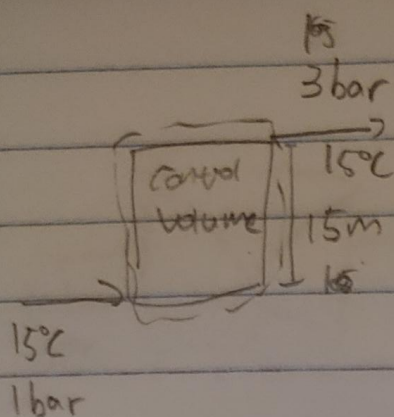
$= \frac{-60 \text{ kJ}}{\text{kg}} - (-55.95 \frac{\text{kJ}}{\text{s}})$

$= \frac{-60 \text{ kJ}}{\text{kg}} \times \frac{1.229 \text{ kg}}{\text{s}} + 55.95 \frac{\text{kJ}}{\text{s}} = -17.79 \frac{\text{kJ}}{\text{s}} = -17.79 \text{ kW}$

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- 0 pts Correct
- ✓ - 3 pts T-V figure
- ✓ - 1 pts T-V figure, $v_1 = ?$ $v_2 = ?$
- 5 pts Q calculation incorrect
- 3 pts scheme incorrect, you mention $\Delta PE = 0$ but your input and output is not on the same level
- 1 pts T-V figure where is step 1 and 2
- 3 pts scheme incorrect. input? output?
- 5 pts m incorrect
- 5 pts scheme

5.



$$m = 1.5 \text{ kg/s}$$

model

- steady state - $\dot{Q} = 0$
- control volume - $\dot{SKE} = 0$
- $m_i = m_e = m$
- 1 inlet & outlet

$$0 = m[(h_1 - h_2) + g(z_1 - z_2)] - \dot{W}$$

$$\nu_f = 1.0009 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$\dot{W} = m[\nu_f(P_1 - P_2) + g(z_1 - z_2)]$$

$$h_1 - h_2 = \nu_f(P_1 - P_2)$$

$$= 1.5 \frac{\text{kg}}{\text{s}} \left[1.0009 \frac{\text{m}^3}{\text{kg}} (100 - 300) + 9.81(-15) \right]$$

$$= -24520.995 \text{ W}$$

$$= -0.521 \text{ kW}$$

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✓ - 0 pts Correct

- 3 pts scheme, storage tank that is 15 m above
reservoir

- 3 pts storage tank that is 15 m above
reservoir; the pump and the storage tank should be on the same level

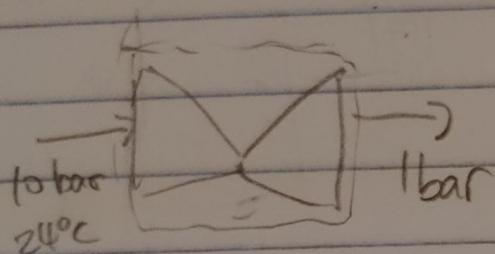
- 5 pts w calculation incorrect

- 3 pts unit in KW

- 3 pts $h_1 = ?$ $h_2 = ?$

- 3 pts engineering model

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model

- Control volume - steady state

- throttling system - 1 inlet & outlet

$$0 = m(h_1 - h_2) \Rightarrow h_1 = h_2$$

$$h_1 = 297.76 \text{ kJ/kg}$$

$$h_2 = 297.76 \text{ kJ/kg} = h_f(1-x) + x h_g$$

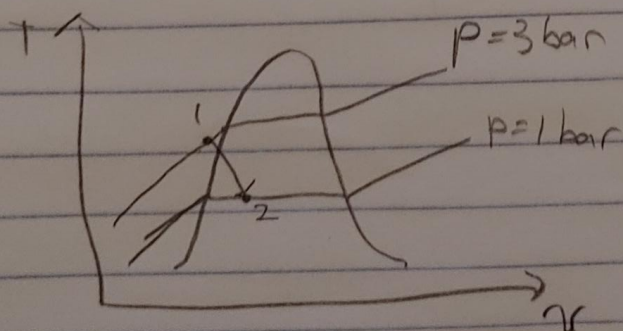
$$h_f = 28.18$$

$$h_g = 1398.41$$

$$\Rightarrow h_2 = h_f - h_f x + x h_g$$

$$h_2 - h_f = x(h_g - h_f)$$

$$\frac{h_2 - h_f}{h_g - h_f} = x = 0.197$$



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